

# Inflation, Information, and Delayed Repricing

Inflacija, informacije i odgođene promjene cijena

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## Abstract

Studies of price rigidity observe prices but rarely show whether a price setter noticed the shock. This paper uses dated media coverage to study inflation and repricing in Croatia's religious sector, for which no official price series exists. Coding identifies 520 posts that connect religion to domestic inflation. Direct sector speech about its own costs or revenues appears in 91 posts and peaks in October 2022. Repricing coverage peaks 20 months later, in June 2024, when the harmonised price level was 25.7 per cent above June 2021. Four named bodies appear in both own-position speech and later repricing coverage; three lags exceed one year. Relevant information was therefore publicly present before adjustment became concentrated. The evidence is consistent with fairness constraints and slow organisational decision-making, but does not identify their separate effects.

**Keywords:** price rigidity; rational inattention; fairness; nonprofit pricing; media as data; Croatia

## Sažetak

Istraživanja rigidnosti cijena promatraju cijene, ali rijetko pokazuju je li određivač cijene uočio šok. Ovaj rad koristi datirano medijsko izvještavanje za analizu inflacije i promjena cijena u hrvatskom vjerskom sektoru, za koji ne postoji službena cjenovna serija. Kodiranjem je utvrđeno 520 objava koje povezuju religiju s domaćom inflacijom. Izravan govor sektora o vlastitim troškovima ili prihodima pojavljuje se u 91 objavi i doseže vrhunac u listopadu 2022. Izvještavanje o promjenama cijena doseže vrhunac 20 mjeseci poslije, u lipnju 2024., kada je harmonizirana razina cijena bila 25,7 posto viša nego u lipnju 2021. Četiri imenovana tijela pojavljuju se i u govoru o vlastitom položaju i u kasnijem izvještavanju o promjeni cijena, a tri odmak dulja su od godine dana. Relevantne su informacije, dakle, bile javno prisutne prije koncentracije prilagodbe. Nalaz je u skladu s ograničenjima pravednosti i sporim organizacijskim odlučivanjem, ali ne razdvaja njihove učinke.

**Ključne riječi:** rigidnost cijena; racionalna nepažnja; pravednost; cijene neprofitnih organizacija; mediji kao podatak; Hrvatska

## 1. Introduction

A fixed price can reflect two different situations. A seller may not have processed new information about costs or demand. Alternatively, the seller may know that the current price is too low but delay adjustment because changing it is costly, slow, or likely to upset customers. The observed price is the same, but the economic mechanism is different.

This distinction is central to the literature on price rigidity. Sticky-information and rational-inattention models emphasise the cost of acquiring and processing information (Mankiw & Reis, 2002; Reis, 2006; Sims, 2003). Menu-cost models assume that sellers know their state but adjust only when the expected gain exceeds a fixed cost (Golosov & Lucas, 2007). Fairness models add the buyer's view of what constitutes a justified price increase (Kahneman et al., 1986). Standard price data rarely show which mechanism is operating.

This paper studies Croatia's religious sector, which provides marriages, funerals, masses for intentions, pilgrimages, charitable services and related activities. Users pay fees or make expected contributions, but no official index records these prices. Media coverage provides a second source of evidence: dated statements about household hardship, institutional costs and revenues, and changes in the price of religious services.

The analysis treats media coverage as an event register rather than a price series. It asks whether the sector publicly discussed its own economic position before repricing coverage became concentrated. It then asks whether the same named bodies appear in both types of event. The design identifies dates, composition, concentration and lags. It does not estimate price levels, pass-through or a structural adjustment hazard.

The main result is a clear ordering. Direct sector speech about its own costs or revenues peaks in October 2022. Repricing coverage peaks in June 2024, 20 months later. Four named bodies appear in both own-position speech and later repricing coverage, and three lags exceed one year. These findings make inattention to the relevant economic object an incomplete explanation. Fairness, governance and adjustment costs remain possible, and the present data cannot separate them.

The paper contributes in three ways. First, it observes public information and delayed repricing in a sector without conventional price microdata. Second, it studies a setting in which nonprofit form, dispersed authority and close user relationships may slow adjustment. Third, it shows why connection coding matters for text-based economic measurement: of 1 450 plausible co-occurrences, 520 are genuine domestic religion-inflation connections.

## **2. Theoretical framework and institutional setting**

### **2.1 Information and price adjustment**

Sticky-information models assume that agents update plans intermittently, so a price can be based on coherent but outdated information (Mankiw & Reis, 2002; Reis, 2006). Rational inattention treats information-processing capacity as scarce and allows firms to focus on signals that matter most for their own decisions (Sims, 2003; Maćkowiak & Wiederholt, 2009). This distinction matters here. Concern about household living costs does not show that a parish or diocese has assessed its own heating bill, wages or budget.

Menu-cost models start from a different point. The seller knows its economic state but pays a fixed cost to change the price. Larger shocks move more sellers across their adjustment thresholds, so adjustment generally becomes more frequent during high inflation (Golosov & Lucas, 2007). Micro evidence also shows substantial variation across sectors and inflation regimes (Bils & Klenow, 2004; Dhyne et al., 2006; Gagnon, 2009; Nakamura & Steinsson, 2008; Alvarez et al., 2019).

These mechanisms can coexist. A seller may update infrequently and still face a cost of acting after an update. The evidence can nevertheless rule out complete explanations. Public discussion of the sector's own costs before later repricing weakens pure inattention to the relevant state. It leaves organisational transmission, adjustment costs and norms as competing explanations.

### **2.2 Fairness and public information**

Firm surveys show that limited information and concern about customer reactions both contribute to slow price adjustment (Blinder et al., 1998; Coibion et al., 2018). Public statements add a contemporaneous measure. They are dated and were made without a researcher's prompt, although they may also be strategic.

The content of the statement therefore matters. General concern for households may reveal little about an institution's pricing decision. A statement about energy bills, wages, repairs or insufficient revenue is closer to the state variable relevant for repricing. It shows that information about the institution's own position was present in public sector speech.

Fairness can still delay action after the shock is known. Under dual entitlement, customers retain a reference transaction while sellers retain a reference return. A cost increase can justify a higher price, but customers may resist increases that appear opportunistic (Kahneman et al., 1986). When inflation is highly visible, the reputational cost of adjustment may also

be high. A seller may then wait even as the accumulated price gap grows. Slow annual budgets or hierarchical approval can produce the same timing, so the analysis tests whether the evidence is consistent with fairness; it does not identify a separate causal effect.

## 2.3 Why the religious sector is informative

Nonprofit organisations cannot distribute residual earnings to owners, often cross-subsidise services, and may include mission directly in their objectives (Hansmann, 1980; Steinberg, 2006). Nonprofit form does not eliminate sluggish or strategic pricing (Jackson, 2014), but it can weaken the usual link between cost and price.

Religious services provide a demanding case. A funeral or mass intention is not viewed as an ordinary market service even when a customary fee exists. There may be no close substitute, fees are not automatically indexed, and decision rights are spread across parishes, dioceses, religious orders and the bishops' conference. These features can slow both information transmission and formal approval.

Catholic social teaching also makes household hardship and fairness part of the sector's public mission. The paper does not evaluate doctrine. It uses that public commitment to motivate the distinction between speech about households and speech about the sector's own economic position. This extends work that treats religious organisations as clubs or welfare providers by examining them as price setters (Iannaccone, 1998; Kallunki & Zrinščak, 2021; Stubbs & Zrinščak, 2009; van Kersbergen & Manow, 2009).

## 3. Data and methods

### 3.1 Corpus and event selection

The DigiKat corpus contains 710 307 Croatian digital media posts published from January 2021 to June 2026. It includes web portals, social platforms and forums, as well as confessional and secular outlets. The target population is religion-salient media content, so the results are not extrapolated to Croatian media as a whole. All posts in the analysis were harmonised and processed with the same selection and coding rules. The 2026 data end in June.

A fixed vocabulary identifies literal references to inflation, rising prices, the cost of living and purchasing power. The general Croatian word for price is excluded because it produces many devotional false positives. A metaphor rule also removes expressions such as "inflation of words" while retaining posts that discuss actual prices. This step identifies 8 041 posts, or 1.13 per cent of the corpus.

The procedure then selects posts in which religious language appears within 220 characters of a cost-of-living expression. This broad rule creates a candidate set; it does not itself establish a substantive connection. Common homonyms and collisions are removed before coding.

The fixed coding pool contains 1 450 posts. Three independent model-based coding runs assessed whether a post concerned literal prices, whether religion was substantively connected, whether the inflation was domestic or foreign, and which subject dominated. Majority decisions identify 652 genuine connections. Of these, 132 concern another country and 520 concern Croatian inflation. The measured set is 0.07 per cent of the corpus and 6.5 per cent of all cost-of-living posts.

**Table 1. From the whole corpus to the set of posts actually measured.**

| Stage                                           | Posts   | % of corpus |
|-------------------------------------------------|---------|-------------|
| All posts in the corpus                         | 710 307 | 100.00      |
| Mention the cost of living                      | 8 041   | 1.13        |
| Selected for coding, religion near the mention  | 1 450   | 0.20        |
| Coded as a genuine connection                   | 652     | 0.09        |
| ... about another country's inflation           | 132     | 0.02        |
| ... about Croatian inflation (the measured set) | 520     | 0.07        |

Source: author's calculations on the DigiKat corpus of 710 307 Croatian digital media posts, January 2021 to June 2026. Posts are selected by a keyword filter and then read and coded one by one; only 45% of the posts the filter selects turn out to connect religion to prices at all. The first two rows are produced by the current code. The set of 1 450 posts sent for coding was fixed when the coding was done and is carried forward unchanged, so the third row is that fixed set rather than a fresh selection; the appendix reports how closely a fresh selection reproduces it.

Only 45 per cent of candidates survive the connection decision. A proximity count alone would therefore confuse topical adjacency with genuine economic content. Because the event is rare, this distinction materially changes the result.

### 3.2 Measures

The first coding scheme distinguishes seven subjects: the price of religious services, the sector as an economic actor, charitable relief, claims about who bears the burden, devotional material, other material and unresolved cases. Four subjects represent observable responses. Posts about the institution as an economic actor form *public voice*; posts about service prices form *repricing*; charity posts form *charitable response*; and burden-sharing claims form *normative response*.

**Table 2. How the religious sector responded, by year.**

| Response            | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 |
|---------------------|------|------|------|------|------|------|
| Public voice        | 8    | 106  | 37   | 9    | 16   | 3    |
| Repricing           | 0    | 46   | 34   | 61   | 13   | 40   |
| Charitable response | 0    | 34   | 11   | 12   | 26   | 4    |
| Normative response  | 2    | 1    | 5    | 4    | 2    | 1    |

Source: author's calculations on the DigiKat corpus of 710 307 Croatian digital media posts, January 2021 to June 2026. The 2026 column covers January to June.

Table 2 describes the timing and composition of the four responses. Public voice is most frequent in 2022, repricing in 2024, charitable response in 2022 and normative response in 2023. The 2026 column covers six months.

Table 3 compares public voice and repricing during 2021–2023, the period in which the main inflation shock occurred.

**Table 3. Public voice and repricing during the main inflation shock.**

| Year | Speaking out | Repricing | Repricing as % of the two |
|------|--------------|-----------|---------------------------|
| 2021 | 8            | 0         | 0.0                       |
| 2022 | 106          | 46        | 30.3                      |
| 2023 | 37           | 34        | 47.9                      |

Source: author's calculations on the DigiKat corpus of 710 307 Croatian digital media posts, January 2021 to June 2026, restricted to 2021–2023.

The repricing share rises from 30.3 per cent in 2022 to 47.9 per cent in 2023. The underlying counts are 106 public-voice and 46 repricing posts in 2022, compared with 37 and 34 in 2023. Coverage therefore begins to move from speech toward repricing before the late 2024 concentration.

**Table 4. What the posts are about, and who published them.**

| Subject of the post             | Posts | %    | Secular outlet | Catholic outlet | Business press |
|---------------------------------|-------|------|----------------|-----------------|----------------|
| The price of religious services | 194   | 37.3 | 189            | 4               | 1              |
| The sector as an economic actor | 179   | 34.4 | 147            | 32              | 0              |
| Charitable relief               | 87    | 16.7 | 58             | 27              | 2              |
| Devotional                      | 26    | 5.0  | 22             | 4               | 0              |
| Who bears the burden            | 15    | 2.9  | 10             | 5               | 0              |
| Unresolved                      | 12    | 2.3  | 11             | 1               | 0              |
| Other                           | 7     | 1.3  | 5              | 2               | 0              |

Source: author's calculations on the DigiKat corpus of 710 307 Croatian digital media posts, January 2021 to June 2026, the 520 posts about Croatian inflation in which religion is genuinely involved. Deciding which of these a post is about was the least reliable of the four coding judgements, so the two largest categories should be read together rather than ranked against each other.

The price of religious services and the sector as an economic actor together account for 373 posts, or 72 per cent of the measured set. Secular outlets report 189 of 194 repricing posts. Across the full set, secular outlets publish 442 posts, or 85 per cent, and Catholic outlets publish 75, or 14 per cent. These figures describe the observation instrument; they do not estimate the behaviour of all outlets.

### 3.3 External price anchor and robustness

Annual Croatian harmonised index of consumer prices (HICP) rates come from Eurostat's `prc_hicp_minr` table. The series covers all items, food and non-alcoholic beverages, and energy from January 2021 to June 2026. All-items inflation peaks at 12.9 per cent in November 2022.

An event register should respond to the underlying economic shock. The monthly share of corpus posts mentioning the cost of living is therefore compared with HICP. Results are shown for an early window containing the inflation surge and a later window with a narrower inflation range. This check follows research on media, inflation expectations and attention thresholds (Čižmešija et al., 2017; Lamla & Maag, 2012; Pfäuti, 2024; Korenok et al., 2026; Aarab et al., 2025).

**Table 5. Does coverage of the cost of living track actual prices?**

| Period             | Months covered     | N  | Inflation ranged from | All items | Food  | Energy | Above vs below 4% |
|--------------------|--------------------|----|-----------------------|-----------|-------|--------|-------------------|
| 2021-01 to 2024-06 | 2021-01 to 2024-06 | 37 | 0.0 to 12.9           | 0.75      | 0.73  | 0.43   | 2.25              |
| 2024-07 to 2026-06 | 2024-07 to 2026-06 | 24 | 3.0 to 5.4            | 0.32      | -0.52 | 0.67   | 0.98              |

Source: author's calculations; Croatian HICP annual rate of change from Eurostat (prc\_hicp\_minr, all items, food and non-alcoholic beverages, energy), retrieved 5 August 2026. Figures are correlations between the monthly share of corpus posts mentioning the cost of living and each price series. The last column is mean coverage in months when inflation was at or above 4% divided by mean coverage below it.

The early window contains 37 observed months. Coverage correlates 0.75 with all-items HICP, 0.73 with food and 0.43 with energy. Mean coverage is 0.51 per cent below 4 per cent inflation and 1.15 per cent at or above it, a ratio of 2.25. Monthly volume ranges from 4 970 to 12 860 posts.

The later window contains 24 months. The corresponding correlations are 0.32, -0.52 and 0.67. Headline inflation varies only from 3.0 to 5.4 per cent, with a standard deviation of 0.60, compared with 4.03 in the early window. Its threshold ratio is 0.98: mean coverage is 1.26 per cent below 4 per cent and 1.24 per cent above it. The narrow range makes this period less informative about an attention threshold.

**Table 6. The same relationship under three different specifications.**

| Period             | Specification                                                | N  | Estimate | Standard error | t    |
|--------------------|--------------------------------------------------------------|----|----------|----------------|------|
| 2021-01 to 2024-06 | Levels, with standard errors robust to trend and persistence | 37 | 0.0914   | 0.0102         | 8.94 |
| 2021-01 to 2024-06 | Month-on-month changes only                                  | 35 | 0.0384   | 0.0895         | 0.43 |
| 2021-01 to 2024-06 | Post counts, allowing for how much was collected             | 37 | 0.1027   | 0.0145         | 7.06 |
| 2024-07 to 2026-06 | Levels, with standard errors robust to trend and persistence | 24 | 0.2879   | 0.1940         | 1.48 |
| 2024-07 to 2026-06 | Month-on-month changes only                                  | 23 | 0.1437   | 0.2371         | 0.61 |
| 2024-07 to 2026-06 | Post counts, allowing for how much was collected             | 24 | 0.2205   | 0.1300         | 1.70 |

Source: author's calculations. The dependent variable is the monthly share of corpus posts mentioning the cost of living, except in the third specification, where it is the monthly count with the log of total posts entered as an offset. Month-on-month changes use consecutive observed months only.

In the early window, the levels estimate remains precise with standard errors robust to trend and persistence ( $t = 8.94$ ). The count model with a volume offset gives a similar result ( $t = 7.06$ ), while first differences are weak ( $t = 0.43$ ). The

corresponding later-window statistics are 1.48, 1.70 and 0.61. The evidence supports a long-cycle relationship between inflation and coverage, not a mechanical month-to-month response.

### 3.4 Coding reliability and additional measures

The original coding was assessed on 173 posts recoded blind with the same written protocol.

**Table 7. Agreement between the original coding and an independent recoding.**

| Judgement                      | Items | Agreement | Kappa |
|--------------------------------|-------|-----------|-------|
| is it about the cost of living | 173   | 0.931     | 0.299 |
| is religion genuinely linked   | 166   | 0.892     | 0.694 |
| domestic or foreign inflation  | 135   | 0.941     | 0.783 |
| what the post is about         | 122   | 0.656     | 0.577 |

Source: author's calculations on a stratified sample of 173 posts independently recoded from the same written protocol with the original labels withheld.

Agreement is 0.931 for literal cost-of-living content, 0.892 for genuine religious linkage, 0.941 for domestic versus foreign inflation and 0.656 for the main subject. The corresponding kappa values are 0.299, 0.694, 0.783 and 0.577. The main subject is the least stable axis, so small adjacent categories are not compared. The author also inspected the coded material during analysis and found no systematic issue that would change the substantive conclusions.

For the present analysis, the 179 public-voice posts receive two additional labels. *Object* records whether the content concerns the sector's own costs or revenues, household hardship, both, or another economic subject. *Voice* records whether a sector actor speaks directly. All 520 posts also receive an institutional-unit label: parish, diocese, religious order, Caritas, bishops' conference, Vatican, church in general, or no acting church unit.

Three blind coding runs applied this extension independently. Pairwise agreement is 0.940 for object, 0.948 for voice and 0.943 for unit. Fleiss' kappa is 0.891, 0.646 and 0.929. Every object and voice item has a majority decision. One unit tie was resolved from the written rule.

Monthly coverage is analysed as an event process. Public voice, own-position speech, HICP and repricing peaks are dated separately. The cumulative HICP price-level gap is reconstructed by chaining the annual rate for the same calendar month from 2021 onward. Named-unit matching requires the same normalised body to appear first in direct own-position speech and later in a repricing report. Duplicate reports of the same body, date and action are collapsed.

## 4. Results

### 4.1 The sector discussed its own economic position

The central test concerns the object of attention. Speech about household hardship does not show that an institution noticed pressure on its own relative price. Speech about heating, wages, repairs, revenue or finances does.

**Table 8. What the institution-register material attends to, and who speaks.**



| Object of economic content               | Sector voice | Outside report | Posts | % of 179 |
|------------------------------------------|--------------|----------------|-------|----------|
| The sector's own costs or revenue        | 88           | 6              | 94    | 52.5     |
| Household hardship                       | 73           | 2              | 75    | 41.9     |
| Both own position and household hardship | 3            | 3              | 6     | 3.4      |
| Other economic subject                   | 2            | 2              | 4     | 2.2      |

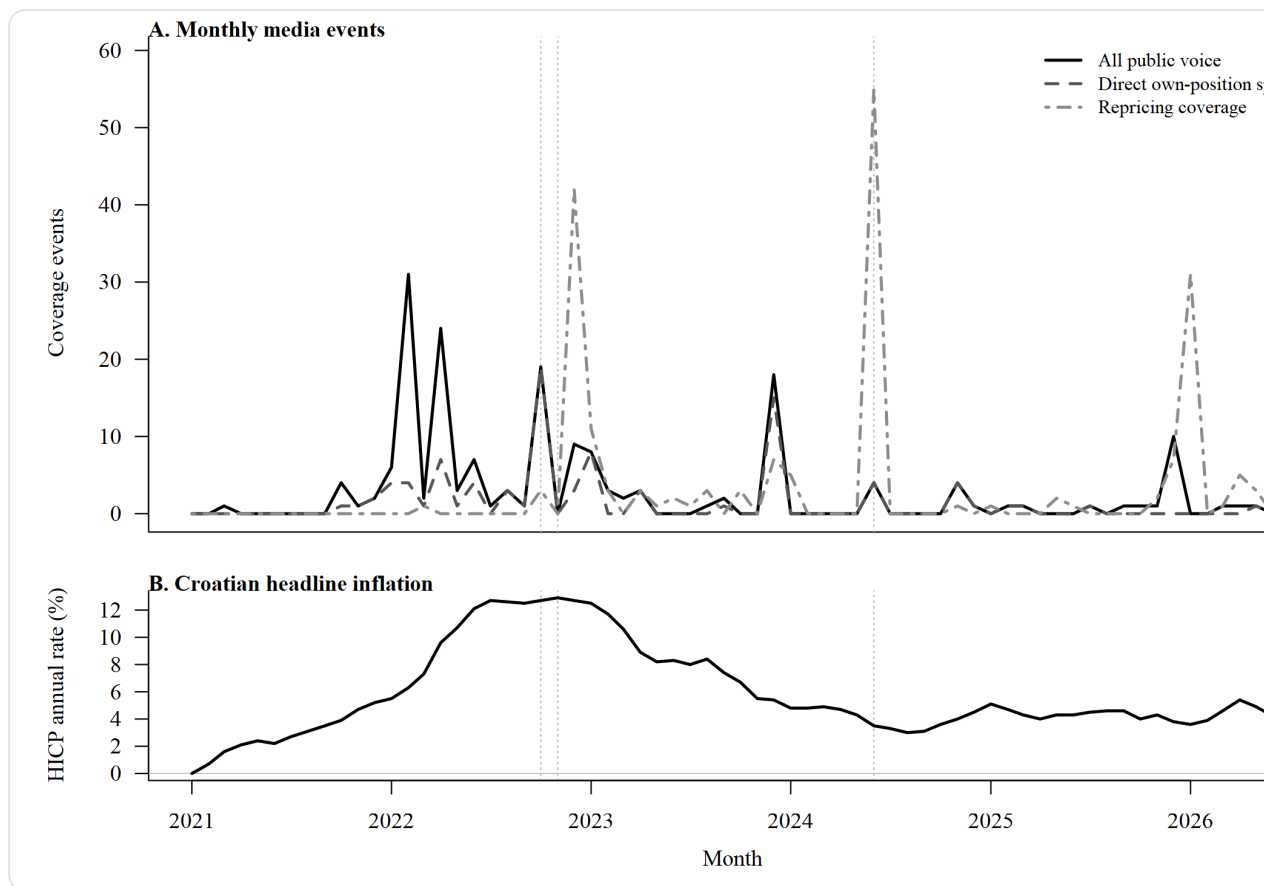
Source: author's recoding of the 179 posts previously classified as the sector acting or speaking in economic coverage. Labels are three-model majorities. A sector voice is a quotation, interview, homily, official statement or press release; an outside report contains no sector actor speaking.

The sector's own costs or revenues are the main object in 94 of 179 public-voice posts. Household hardship appears in 75 posts, both objects in 6 and another economic subject in 4. A sector actor speaks directly in 88 own-position posts and 3 both-object posts. This gives 91 direct own-position events, or 50.8 per cent of the set. Direct household-hardship speech appears in 76 posts.

Household-hardship speech peaks in February 2022 with 27 posts. Own-position speech peaks eight months later, in October 2022, with 19 posts. This is one month before the November 2022 HICP peak and 20 months before the repricing-coverage peak. Of the 91 direct own-position posts, 78 occur before June 2024.

**Figure 1. Public economic speech and repricing coverage, January 2021–June 2026.**





Monthly public voice, direct own-position speech and repricing coverage

Source: author's calculations. Lines show monthly coverage events. Key dates are the public-voice and household peaks (2022-02), first repricing coverage (2022-03), own-position peak (2022-10), HICP peak (2022-11) and repricing peak (2024-06). The figure reports media coverage, not an official fee index. The 2026 series ends in June.

Relevant information about the sector's own position was therefore publicly present before repricing coverage became concentrated. This rules out exclusive attention to household hardship as an aggregate explanation. It does not show that every price-setting unit received the same information or that public statements reproduce private beliefs.

## 4.2 Information and repricing in named bodies

The unit analysis tests whether information and repricing were confined to different parts of the organisation. Public voice is concentrated in dioceses and the bishops' conference, while repricing is spread across the conference, dioceses and parishes.

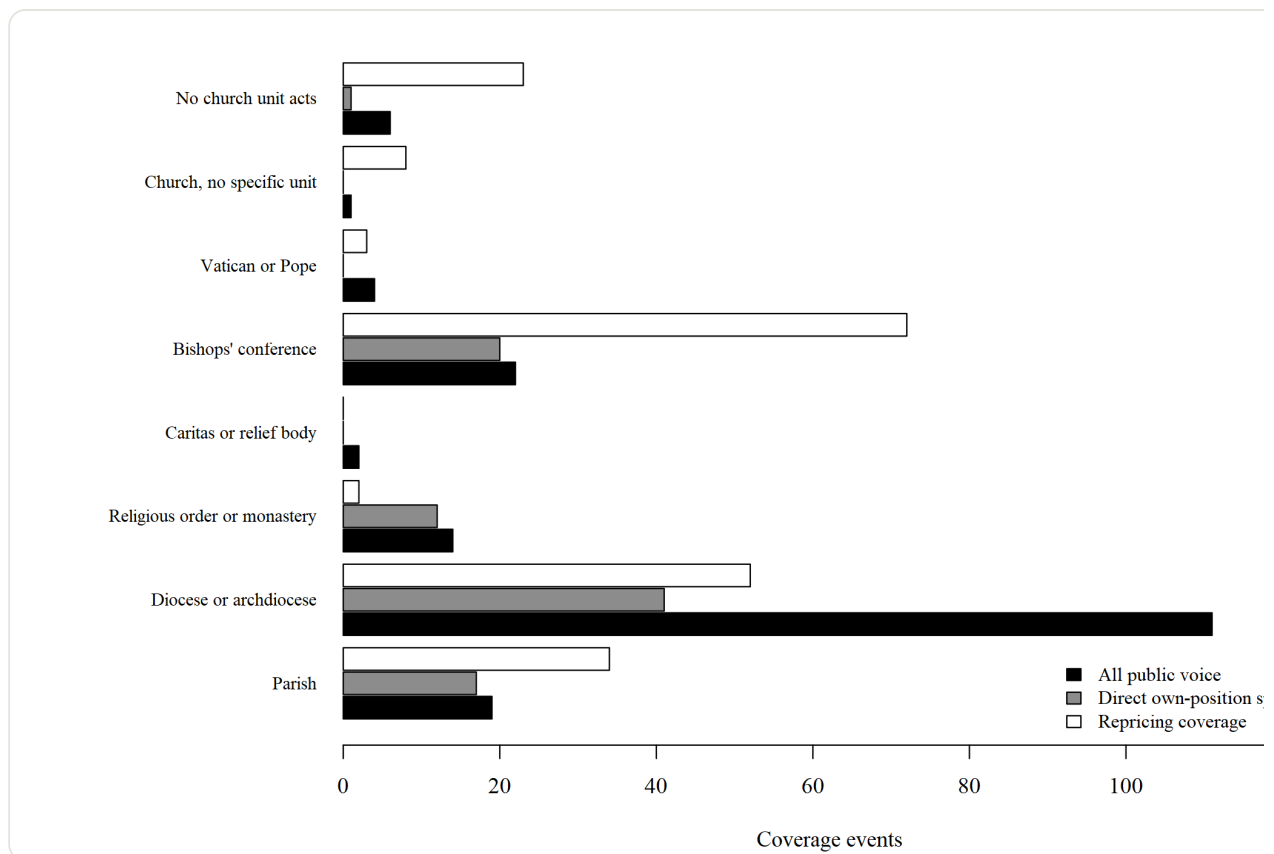
**Table 9. Where speech and adjustment are located inside the sector.**

| Institutional unit           | Direct own-position speech | All public voice | Repricing | Charitable response | Matched |
|------------------------------|----------------------------|------------------|-----------|---------------------|---------|
| Parish                       | 17                         | 19               | 34        | 2                   |         |
| Diocese or archdiocese       | 41                         | 111              | 52        | 0                   |         |
| Religious order or monastery | 12                         | 14               | 2         | 0                   |         |
| Caritas or relief body       | 0                          | 2                | 0         | 78                  |         |
| Bishops' conference          | 20                         | 22               | 72        | 0                   |         |
| Vatican or Pope              | 0                          | 4                | 3         | 1                   |         |
| Church, no specific unit     | 0                          | 1                | 8         | 0                   |         |
| No church unit acts          | 1                          | 6                | 23        | 6                   |         |

Source: author's recoding of all 520 posts. Named-unit matches require the same specific unit to speak about its own position before a later repricing report; identifying names remain in the private analysis files. Counts are coverage events, not a census of institutional actions.

Dioceses account for 111 public-voice and 52 repricing reports. The bishops' conference accounts for 22 and 72, while parishes account for 19 and 34. Religious orders account for 14 and 2. Caritas dominates charitable response with 78 posts and does not appear as a repricing unit.

**Figure 2. Public voice and repricing coverage by institutional unit.**



Distribution of public voice and repricing coverage across institutional units

Source: author's calculations on the 520-post measured set. Bars represent media coverage events, not a census of institutional actions.

At least one relevant event identifies 32 specific units. Of these, 22 appear in direct own-position speech and 14 in repricing coverage. Four named bodies appear in both, with speech first. Their lags range from 1 to 1 505 days and have a median of 669 days. Three matches exceed one year. The matched bodies are three diocesan units and one bishops' conference.

The one-day match represents a contemporaneous discussion and adjustment. The other three provide stronger within-body sequences. They do not show that the same individual made both decisions, but they show that organisational separation cannot explain every observed delay.

### 4.3 Timing of repricing coverage

The register first records repricing coverage in March 2022, so the claim is not that every fee remained unchanged until 2024. The claim concerns timing and concentration. Public voice peaks in February 2022, own-position speech in October 2022, HICP in November 2022 and repricing coverage in June 2024. The last peak occurs 28 months after the overall public-voice peak and 19 months after the HICP peak.

June 2024 contains 55 repricing posts. This is 28.4 per cent of all repricing coverage and 90.2 per cent of the 2024 repricing material. In that month, the chained HICP level is 25.7 per cent above June 2021.

Two checks reduce the risk that this pattern reflects reporting capacity alone. The register already captures 46 repricing posts in 2022 and 34 in 2023, so it could detect repricing before the peak. Secular outlets publish 189 of 194 repricing reports and had a strong incentive to report church price increases when inflation was most salient.

These checks cannot recover unreported fee schedules or determine the date on which a fee formally changed. The result is therefore stated as the timing of

repricing coverage

. Direct diocesan and parish fee schedules are needed to convert it into a price-duration estimate.

#### **4.4 Interpretation: fairness and organisational delay**

The ordering is consistent with a fairness constraint. Household hardship becomes salient first, the sector then discusses its own costs near the inflation peak, and repricing coverage becomes concentrated after inflation and public voice have fallen. In a mission-based relationship, adjustment after public salience declines may be less costly than adjustment at the peak.

A fixed technical menu cost is unlikely to be the whole explanation. The HICP price level rose 25.7 per cent, and the sector publicly discussed the relevant information well before the repricing peak. This does not reject menu costs; it shows that another friction is also needed.

Organisational delay can produce the same ordering. Fee changes may require diocesan consultation, annual budgeting or coordination between national and local bodies. The data contain no variable that changes fairness while holding organisational cost fixed. The most defensible interpretation is therefore joint: the pattern is consistent with fairness constraints and slow governance, but their separate effects are not identified.

#### **4.5 Public narratives and charitable response**

The distributional category supplies a secondary result about public economic narratives. Claims that identify who bears the burden account for 15 of 520 posts, or 2.9 per cent. The annual counts are 2, 1, 5, 4, 2 and 1. A broader reading that includes adjacent structural language reaches at most 8 per cent.

This does not imply an absence of relief. Charitable response accounts for 87 posts, or 16.7 per cent, and appears throughout the period. Rather, structural explanations are scarce compared with coverage of pricing, institutional finances and relief. In the terms of narrative economics, the available stories organise how economic problems are understood (Shiller, 2017).

The result applies to the national digital record. News coverage may miss homilies, pastoral letters and local action, so this finding remains secondary to the price-adjustment result.

#### **4.6 Why connection coding matters**

Text-based economic indicators often begin from document frequency. Baker et al. (2016) demonstrate the value of newspaper counts when the concept and dictionary are carefully validated. Gentzkow et al. (2019) and Grimmer and Stewart (2013) emphasise that text measurements remain problem-specific and require validation against the intended construct.

The cost-of-living vocabulary identifies 8 041 posts, and proximity to religious language selects 1 450. Only 652 contain a genuine connection, of which 520 concern Croatia. Long articles may mention the two subjects in separate passages, religious words may have secular meanings, and news feeds may combine unrelated items.

Without connection coding, the timing analysis would mix genuine economic events with topical adjacency. Precision is therefore part of the economic measurement, not a cosmetic validation statistic.

### **5. Discussion**

#### **5.1 What the event register establishes**

The event register does not reveal private cognition, but it adds information that a price series cannot provide. Direct sector speech about its own economic position precedes the repricing concentration by 20 months. Four named bodies show the same sequence, and three lags exceed one year.

Attention to household hardship alone cannot explain the aggregate pattern because own-position speech is slightly more common. Organisational separation remains important: parishes, dioceses and the bishops' conference have different

decision rights, and most events cannot be matched to the same named body. The four matches nevertheless show that separation is not a complete explanation.

Once information about the sector's own position is publicly present, delayed action requires an additional friction. Fairness, governance and adjustment costs become part of the explanation. The analysis does not estimate their macroeconomic importance, but it narrows the set of plausible mechanisms.

## **5.2 A boundary condition for price adjustment**

Most evidence on price duration comes from competitive retail goods and services (Dhyne et al., 2006; Klenow & Malin, 2010; Nakamura & Steinsson, 2008). The religious sector differs through weak competitive pressure, no automatic indexation, dispersed authority and relationships partly governed by norms. A long and concentrated adjustment does not contradict state-dependent pricing in general. It identifies a setting in which the usual positive link between inflation and adjustment frequency may weaken.

The accumulated gap is economically meaningful even without a sector price index. By June 2024, the HICP level was 25.7 per cent above June 2021. The paper cannot measure the sector's contribution to aggregate price dispersion, but it identifies a setting where relative-price adjustment may be large and statistically invisible.

Croatia adopted the euro in January 2023, which may have created an occasion for some fee changes. Aggregate evidence finds no robust overall conversion effect and locates the main effects in selected goods and services (Falagiarda et al., 2023; Sorić, 2024). The observed sequence begins before conversion and continues long after it, so the euro changeover cannot explain the full October 2022–June 2024 interval.

## **5.3 What media can and cannot measure**

Media can provide an observation instrument for sectors outside official statistics. Similar gaps arise for informal providers, clubs, small family firms and organisations below reporting thresholds. A dated corpus can recover reported events and contemporaneous statements where administrative data do not exist.

Coverage is nevertheless shaped by newsworthiness. Inflation can affect both the underlying event rate and the probability of reporting. A bishop's statement is easier to observe than a parish price list, and coverage may follow rather than coincide with a formal decision.

The estimand must therefore remain narrow. The paper reports coverage events, dated peaks, chained aggregate price gaps and matched-unit sequences. It does not estimate a fee level, pass-through coefficient or structural hazard. Direct fee schedules would permit those stronger tests.

## **5.4 Limitations**

Four limits define the claim. First, public speech can be strategic and does not certify private cognition. Second, named-unit matching is sparse: four matches cannot represent a sector with many local units. Third, coverage can lag the underlying price decision, and only external fee schedules can measure that lag. Fourth, the design cannot separate fairness from organisational decision costs.

These limits narrow the causal interpretation, but standard price data would not show the object of public attention or its organisational location. The event register makes those distinctions visible and identifies the data needed for a stronger test.

## **6. Conclusion**

A fixed price cannot show whether a seller failed to notice a shock or noticed it and delayed action. The Croatian religious sector provides a second observation channel through public speech and repricing coverage.

The sequence is clear. Household-hardship speech peaks in February 2022, own-position speech in October 2022 and repricing coverage in June 2024. By the last date, the HICP price level was 25.7 per cent above June 2021. Four named bodies appear in both own-position speech and later repricing coverage, and three lags exceed one year.

Relevant public information was therefore present before repricing coverage became concentrated. The pattern is consistent with fairness constraints operating through a slow hierarchy. Pure inattention to the relevant object and a fixed technical menu cost are incomplete explanations, although the data cannot separate fairness from organisational delay.

The next step is to recover dated fee schedules from dioceses and parishes. Combining those prices with the public-information record would distinguish reporting delay from decision delay and allow direct estimates of price duration.

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## Appendix A. Reproduction

The study runs from the repository root as a numbered R pipeline. Script 00 retrieves and validates Eurostat HICP. Scripts 01–03 reconstruct the cost-of-living tag, candidate pool and measured set. Scripts 04–07 produce the sector profile, instrument checks and independent recoding. Scripts 14–16 create the additional labels, unit matches and event-timing measures. Script 17 generates the figures, and Script 18 publishes the HTML, PDF and Word editions.

Script 08 with `--v2` generates all nine manuscript tables and the scalar checks used in the text. Script 09 installs the table fragments, and Script 10 checks the tables, derived values, abstracts and encoding. Script 11 renders full and blinded Word files outside the repository. Script 12 assembles a disclosure-screened replication package. The end-to-end command is `Rscript studies/inflation-salience/RUN_ALL.R --v2 --no-network`.

The original candidate-selection scripts were lost and reconstructed from the written specification. The reconstructed measured set reproduces all 520 final items and all seven Table 4 categories. A fresh proximity selection recovers 94.5 per cent of the fixed 1 450 coding pool while adding candidates of its own. No result reselects or changes the fixed pool.

## Appendix B. Coding protocol

For the original coding, each annotator saw a date, outlet type, headline and an excerpt centred on the cost-of-living phrase. No existing decision and no other annotator's label was visible. The four conditional decisions asked whether the item concerned literal prices, whether religion was genuinely linked rather than incidentally adjacent, whether the inflation was domestic or foreign, and which substantive register dominated. Majority voting across three runs produced the labels.

The version 2 addendum operates only on those already coded posts. For the 179 institution-register items, `object` is `own` when the economic content concerns the institution's heating, energy, wages, maintenance, insurance, building repair, collections, fees, finances, budgets, property or investment. It is `household` when the content concerns the cost of living borne by families, pensioners, households or the poor. `both` requires comparable weight, and `other` covers neither `object`. The decision concerns the object of economic content, not the speaker.

The `voice` label is `sector` when a sector actor speaks through a quotation, interview, homily, official statement or press release. It is `outside` when the sector is written about without a sector actor speaking. Direct evidence for relevant public information is `own` or `both` together with `sector`.

Every one of the 520 posts receives a `unit` label. The unit is the body performing the reported economic action. Allowed values are parish, diocese, religious order, Caritas, bishops' conference, Vatican, church in general and none. The specific name is recorded exactly as written, kept private, normalised conservatively and used only for matching. A named match requires direct own-position sector speech to precede repricing coverage for the same normalised body. Aggregate matched counts leave the private directory; names do not.

```
window.document.addEventListener("DOMContentLoaded", function (event) { const tabsets =
window.document.querySelectorAll(".panel-tabset-tabby") tabsets.forEach(function(tabset) { const tabby = new Tabby('#' +
tabset.id); }); const icon = "☐"; const anchorJS = new window.AnchorJS(); anchorJS.options = { placement: 'right', icon: icon };
anchorJS.add(".anchored"); const isCodeAnnotation = (el) => { for (const clz of el.classList) { if (clz.startsWith('code-
annotation-')) { return true; } } return false; } const onCopySuccess = function(e) { // button target const button = e.trigger; //
don't keep focus button.blur(); // flash "checked" button.classList.add('code-copy-button-checked'); var currentTitle =
button.getAttribute("title"); button.setAttribute("title", "Copied!"); let tooltip; if (window.bootstrap) {
button.setAttribute("data-bs-toggle", "tooltip"); button.setAttribute("data-bs-placement", "left"); button.setAttribute("data-
bs-title", "Copied!"); tooltip = new bootstrap.Tooltip(button, { trigger: "manual", customClass: "code-copy-button-tooltip",
offset: [0, -8]}); tooltip.show(); } setTimeout(function() { if (tooltip) { tooltip.hide(); button.removeAttribute("data-bs-title");
button.removeAttribute("data-bs-toggle"); button.removeAttribute("data-bs-placement"); } button.setAttribute("title",
currentTitle); button.classList.remove('code-copy-button-checked'); }, 1000); // clear code selection e.clearSelection(); }
const getTextToCopy = function(trigger) { const outerScaffold = trigger.parentElement.cloneNode(true); const codeEl =
outerScaffold.querySelector('code'); for (const childEl of codeEl.children) { if (isCodeAnnotation(childEl)) { childEl.remove(); } }
return codeEl.innerText; } const clipboard = new window.ClipboardJS('.code-copy-button:not([data-in-quarto-modal])', {
text: getTextToCopy }); clipboard.on('success', onCopySuccess); if (window.document.getElementById('quarto-embedded-
source-code-modal')) { const clipboardModal = new window.ClipboardJS('.code-copy-button[data-in-quarto-modal]', { text:
getTextToCopy, container: window.document.getElementById('quarto-embedded-source-code-modal') });
clipboardModal.on('success', onCopySuccess); } var localhostRegex = new RegExp(/^(?:http|https):\/\/localhost(?:[0-9]*\/)/);
var mailtoRegex = new RegExp(/^mailto:/); var filterRegex = new RegExp('/') + window.location.host + '/'; var isInternal =
```

```

(href) => { return filterRegex.test(href) || localhostRegex.test(href) || mailtoRegex.test(href); } // Inspect non-navigation links
and adorn them if external var links = window.document.querySelectorAll('a[href]:not(.nav-link):not(.navbar-
brand):not(.toc-action):not(.sidebar-link):not(.sidebar-item-toggle):not(.pagination-link):not(.no-external):not([aria-
hidden]):not(.dropdown-item):not(.quarto-navigation-tool):not(.about-link)'); for (var i=0; i<links.length; i++) { const link =
links[i]; if (!isInternal(link.href)) { // undo the damage that might have been done by quarto-nav.js in the case of // links that
we want to consider external if (link.dataset.originalHref !== undefined) { link.href = link.dataset.originalHref; } } } function
tippyHover(el, contentFn, onTriggerFn, onUntriggerFn) { const config = { allowHTML: true, maxWidth: 500, delay: 100, arrow:
false, appendTo: function(el) { return el.parentElement; }, interactive: true, interactiveBorder: 10, theme: 'light-border',
placement: 'bottom-start', }; if (contentFn) { config.content = contentFn; } if (onTriggerFn) { config.onTrigger = onTriggerFn; } if
(onUntriggerFn) { config.onUntrigger = onUntriggerFn; } window.tippy(el, config); } const noterefs =
window.document.querySelectorAll('a[role="doc-noteref"]'); for (var i=0; i<noterefs.length; i++) { const ref = noterefs[i];
tippyHover(ref, function() { // use id or data attribute instead here let href = ref.getAttribute('data-footnote-href') ||
ref.getAttribute('href'); try { href = new URL(href).hash; } catch {} const id = href.replace(/^#\//, ""); const note =
window.document.getElementById(id); if (note) { return note.innerHTML; } else { return ""; } }); } const xrefs =
window.document.querySelectorAll('a.quarto-xref'); const processXRef = (id, note) => { // Strip column container classes
const stripColumnClz = (el) => { el.classList.remove("page-full", "page-columns"); if (el.children) { for (const child of
el.children) { stripColumnClz(child); } } } stripColumnClz(note) if (id === null || id.startsWith('sec-')) { // Special case sections,
only their first couple elements const container = document.createElement("div"); if (note.children && note.children.length
> 2) { container.appendChild(note.children[0].cloneNode(true)); for (let i = 1; i < note.children.length; i++) { const child =
note.children[i]; if (child.tagName === "P" && child.innerText === "") { continue; } else {
container.appendChild(child.cloneNode(true)); break; } } if (window.Quarto?.typesetMath) {
window.Quarto.typesetMath(container); } return container.innerHTML } else { if (window.Quarto?.typesetMath) {
window.Quarto.typesetMath(note); } return note.innerHTML; } } else { // Remove any anchor links if they are present const
anchorLink = note.querySelector('a.anchorjs-link'); if (anchorLink) { anchorLink.remove(); } if (window.Quarto?.typesetMath)
{ window.Quarto.typesetMath(note); } if (note.classList.contains("callout")) { return note.outerHTML; } else { return
note.innerHTML; } } for (var i=0; i<xrefs.length; i++) { const xref = xrefs[i]; tippyHover(xref, undefined, function(instance) {
instance.disable(); let url = xref.getAttribute('href'); let hash = undefined; if (url.startsWith('#')) { hash = url; } else { try { hash =
new URL(url).hash; } catch {} } if (hash) { const id = hash.replace(/^#\//, ""); const note =
window.document.getElementById(id); if (note !== null) { try { const html = processXRef(id, note.cloneNode(true));
instance.setContent(html); } finally { instance.enable(); instance.show(); } } } else { // See if we can fetch this fetch(url.split('#')
[0]).then(res => res.text()).then(html => { const parser = new DOMParser(); const htmlDoc = parser.parseFromString(html,
"text/html"); const note = htmlDoc.getElementById(id); if (note !== null) { const html = processXRef(id, note);
instance.setContent(html); } }).finally(() => { instance.enable(); instance.show(); } }); } else { // See if we can fetch a full url
(with no hash to target) // This is a special case and we should probably do some content thinning / targeting fetch(url)
.then(res => res.text()).then(html => { const parser = new DOMParser(); const htmlDoc = parser.parseFromString(html,
"text/html"); const note = htmlDoc.querySelector('main.content'); if (note !== null) { // This should only happen for chapter
cross references // (since there is no id in the URL) // remove the first header if (note.children.length > 0 &&
note.children[0].tagName === "HEADER") { note.children[0].remove(); } const html = processXRef(null, note);
instance.setContent(html); } }).finally(() => { instance.enable(); instance.show(); } }); }, function(instance) { }); } let
selectedAnnotEl; const selectorForAnnotation = ( cell, annotation) => { let cellAttr = 'data-code-cell="' + cell + '"'; let lineAttr
= 'data-code-annotation="' + annotation + '"'; const selector = 'span[' + cellAttr + '][' + lineAttr + ']; return selector; } const
selectCodeLines = (annotEl) => { const doc = window.document; const targetCell = annotEl.getAttribute("data-target-cell");
const targetAnnotation = annotEl.getAttribute("data-target-annotation"); const annotSpan =
window.document.querySelector(selectorForAnnotation(targetCell, targetAnnotation)); const lines =
annotSpan.getAttribute("data-code-lines").split(","); const lineIds = lines.map((line) => { return targetCell + "-" + line; }) let
top = null; let height = null; let parent = null; if (lineIds.length > 0) { //compute the position of the single el (top and bottom
and make a div) const el = window.document.getElementById(lineIds[0]); top = el.offsetTop; height = el.offsetHeight; parent
= el.parentElement.parentElement; if (lineIds.length > 1) { const lastEl =
window.document.getElementById(lineIds[lineIds.length - 1]); const bottom = lastEl.offsetTop + lastEl.offsetHeight; height =
bottom - top; } if (top !== null && height !== null && parent !== null) { // cook up a div (if necessary) and position it let div =
window.document.getElementById("code-annotation-line-highlight"); if (div === null) { div =

```

```
'absolute'; parent.appendChild(div); } div.style.top = top - 2 + "px"; div.style.height = height + 4 + "px"; div.style.left = 0; let gutterDiv = window.document.getElementById("code-annotation-line-highlight-gutter"); if (gutterDiv === null) { gutterDiv = window.document.createElement("div"); gutterDiv.setAttribute("id", "code-annotation-line-highlight-gutter"); gutterDiv.style.position = 'absolute'; const codeCell = window.document.getElementById(targetCell); const gutter = codeCell.querySelector('.code-annotation-gutter'); gutter.appendChild(gutterDiv); } gutterDiv.style.top = top - 2 + "px"; gutterDiv.style.height = height + 4 + "px"; } selectedAnnoteEl = annoteEl; }); const unselectCodeLines = () => { const elementsIds = ["code-annotation-line-highlight", "code-annotation-line-highlight-gutter"]; elementsIds.forEach((elId) => { const div = window.document.getElementById(elId); if (div) { div.remove(); }); selectedAnnoteEl = undefined; }; // Handle positioning of the toggle window.addEventListener("resize", throttle(() => { elRect = undefined; if (selectedAnnoteEl) { selectCodeLines(selectedAnnoteEl); }, 10 )); function throttle(fn, ms) { let throttle = false; let timer; return (...args) => { if(!throttle) { // first call gets through fn.apply(this, args); throttle = true; } else { // all the others get throttled if(timer) clearTimeout(timer); // cancel #2 timer = setTimeout(() => { fn.apply(this, args); timer = throttle = false; }, ms); }} }; // Attach click handler to the DT const annotatedIs = window.document.querySelectorAll('dt[data-target-cell]'); for (const annoteDlNode of annotatedIs) { annoteDlNode.addEventListener('click', (event) => { const clickedEl = event.target; if (clickedEl !== selectedAnnoteEl) { unselectCodeLines(); const activeEl = window.document.querySelector('dt[data-target-cell].code-annotation-active'); if (activeEl) { activeEl.classList.remove('code-annotation-active'); } selectCodeLines(clickedEl); clickedEl.classList.add('code-annotation-active'); } else { // Unselect the line unselectCodeLines(); clickedEl.classList.remove('code-annotation-active'); }}} ); const findCites = (el) => { const parentEl = el.parentElement; if (parentEl) { const cites = parentEl.dataset.cites; if (cites) { return [el, cites.split(' ') ]; } else { return findCites(el.parentElement) } } else { return undefined; } }; var bibliorefs = window.document.querySelectorAll('a[role="doc-biblioref"]'); for (var i=0; i<bibliorefs.length; i++) { const ref = bibliorefs[i]; const citeInfo = findCites(ref); if (citeInfo) { tippyHover(citeInfo.el, function() { var popup = window.document.createElement('div'); citeInfo.cites.forEach(function(cite) { var citeDiv = window.document.createElement('div'); citeDiv.classList.add('hanging-indent'); citeDiv.classList.add('csl-entry'); var biblioDiv = window.document.getElementById('ref-' + cite); if (biblioDiv) { citeDiv.innerHTML = biblioDiv.innerHTML; } popup.appendChild(citeDiv); }); return popup.innerHTML; })); }
```